

Proposal for UAV State Estimation and Motion Modeling System

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Section: 0102

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**WHAT ARE YOU PASSIONATE ABOUT, WHAT IS IMPORTANT TO YOU,
AND WHAT DO YOU ENJOY?**

I've always been interested in how software interacts with real physical systems, especially in aerospace applications. I enjoy using programming and data analysis to solve real engineering problems. Lately, I've been drawn to topics like UAV navigation, flight dynamics, and how sensors can be used to estimate a vehicle's motion in uncertain environments, as I feel this is what bridges the gap between what I am learning within my courses and industry. Being able to simulate these systems and then connect the math to real flight data is something I find really rewarding, along with the possibility to deepen my learning into application and to gain valuable skills in doing so.

**RESOURCES OR PEOPLE OUTSIDE OF THIS CLASS/SCHOOL YOU ARE
CONSIDERING USING.**

I plan to use open-source UAV datasets and online simulation tools. Specifically, I will use the AERPAW RF Sensor Measurements with UAV (July 2024) dataset, which provides real UAV positional data (X, Y, Z) and ground-truth flight paths. I may also use resources from research papers that discuss UAV localization and state estimation techniques, as well as Python libraries like NumPy, SciPy, Pandas, pyproj, and Matplotlib. I'll also reference lecture material from ENAE380 labs, especially the ones on Signals, FFT, Filtering, and Path Planning. The dataset also includes information on when the UAV had a clear line of sight or a non-line-of-sight connection with the receiver towers, and it provides time difference of arrival measurements that can be used to analyze signal timing and positioning accuracy.

PROJECT DESCRIPTION.

For my project, I want to build a UAV State Estimation and Motion Modeling System that uses a Kalman Filter to estimate a drone's true position based on simulated and real-

world data. The first part of my project will focus on testing different motion models (like constant velocity and constant acceleration) using simulated 2D position data. Once the filter performs well in simulation, I'll apply it to open-source UAV flight data from the *AERPAW RF Sensor Measurements with UAV (July 2024)* dataset[1] to test how well it can track and smooth noisy real-world positions.

The overall goal for my final project is to develop a Kalman Filter that can de-noise UAV positional data from the AERPAW RF Sensor Measurements with UAV (July 2024) dataset and improve the accuracy of the recorded flight paths. This project will give me hands-on experience with data fusion and filtering—skills that directly support my long-term goal of designing and building a fully functional flight computer after the semester. My future plan is to expand this work to operate across multiple flight platforms, including both fixed-wing aircraft and multirotor UAVs, so that I can eventually integrate my own state estimation and navigation algorithms into real-world flight systems.

NEXT STEPS, SKILLS TO LEARN, AND TOPICS TO RESEARCH.

My next steps will be to finalize the mathematical model for the Kalman Filter and write the core Python functions for the prediction and correction steps. This foundation was developed in the FFT and Kalman Filter lab earlier in the semester, but I plan to do additional research to strengthen my mathematical intuition and optimize the design for this project. After that, I will generate synthetic data using NumPy to test the filter in a controlled environment. This step will rely on motion modeling equations and the addition of noise to simulate real-world data. Once I confirm that the filter performs correctly in simulation, I will move on to processing the AERPAW dataset by extracting the UAV's geographical locations (latitude and longitude) and converting them into X, Y, and Z positions with associated timestamps.

To complete this project, I'll need to review several new topics, including discrete-time state-space modeling, process and measurement noise covariance tuning, and methods for integrating real datasets into Python. I also plan to improve my data visualization and

result presentation skills using Matplotlib—specifically, comparing true versus estimated UAV paths—and practice communicating those results more effectively, similar to how they would be reported in an engineering setting.

For my workflow, I plan to first test and simulate the project within a Jupyter Notebook to document my reasoning and conceptual understanding. After the system is validated, I'll convert the notebook into a standalone Python script that can be further developed and applied after the semester ends. Finally, I will document the entire process and create a short video walkthrough demonstrating how the Kalman Filter performs on both simulated and real UAV data, as outlined in the project rubric.

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- [1] AERPAW Project Team, *AERPAW RF Sensor Measurements with UAV (July 2024)*. North Carolina State University, 2024. Available at <https://aerpaw.org/dataset/aerpaw-rf-sensor-measurements-with-uav-july-2024/>.